PACT-GUI: EFFICIENT GUI AGENTS WITH ACTION-ANCHORED GLOBAL–LOCAL HISTORY

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ABSTRACT

GUI agents for multi-step tasks repeatedly encode high-resolution visual histories, incurring substantial inference costs. While token and KV-cache compression can reduce downstream computation, they leave the cost of visual encoding largely unchanged. However, downsampling historical screenshots can obscure local state changes, while visual redundancy-based pruning may discard seemingly similar regions that remain critical for action prediction. Our analysis shows that GUI histories are globally redundant but locally heterogeneous, and that visually similar content can still affect action prediction. These findings motivate PACT-GUI, a training-free framework that uses executed actions as spatial anchors for history compression. Specifically, before visual encoding, we represent each historical screenshot as a low-resolution global thumbnail paired with a high-resolution crop around the corresponding interaction. To achieve end-to-end acceleration, we pair this representation with template-based structured decoding, which separates the fixed output template from its variable fields, allowing the model to process known syntax in blocks and autoregressively predict only the action type and its parameters. Across four GUI benchmarks, PACT-GUI achieves a $6.62\times$ reduction in mean end-to-end computation, demonstrating substantial efficiency gains.

1 INTRODUCTION

Recent vision-language models (VLMs) have made it practical to ground natural-language intent in rendered interfaces (Alayrac et al., 2022; Li et al., 2023; Dai et al., 2023; Bai et al., 2023; Peng et al., 2023; Liu et al., 2023b;a; Wang et al., 2024). Building on this capability, graphical user interface (GUI) agents translate user goals into grounded actions over successive screens (Rawles et al., 2023; Deng et al., 2023; Li et al., 2024a; Furuta et al., 2024; Zhou et al., 2023; Koh et al., 2024; Drouin et al., 2024; Rawles et al., 2024; Gao et al., 2023; Liu et al., 2023c), providing a common interaction interface for mobile and web environments (Cheng et al., 2024; You et al., 2024; Gou et al., 2025; Wu et al., 2024; Hong et al., 2023; Baechler et al., 2024; Zheng et al., 2024a; He et al., 2024; Xie et al., 2024; Zhang et al., 2025a; Lu et al., 2025). However, high-resolution visual inference over the current screen and interaction history makes this flexibility costly: in our AndroidControl-Curated evaluation, the uncompressed QWEN3-VL-4B-INSTRUCT baseline requires 5.540 s per instance on NVIDIA L20 GPUs on average (Table 2), and such delays compound over multi-step trajectories.

Much of this growing cost comes from repeatedly processing high-resolution visual history. For each processing round, the agent concatenates full-resolution historical screenshots with the current screen; this composite visual input incurs preprocessing, visual-encoding, and multimodal-prefilling costs. Subsequent action generation introduces an independent cost because standard decoding autoregressively emits both task-dependent content and protocol-determined syntax. As the trajectory lengthens, optimizing only the visual context length or only the decoding path therefore yields an incomplete solution. We consequently study the quality–efficiency trade-off throughout the end-to-end generation within this standard setting.

Existing acceleration methods typically optimize individual stages of multi-step inference, leaving a gap in end-to-end efficiency. History-aware GUI agents and efficient VLMs compress trajectories, visual tokens, or KV states after visual content is encoded (Chen et al., 2025; Zhang et al., 2025a; Li et al., 2026; Huang et al., 2025; Zhou et al., 2026a); they reduce downstream computation but retain the cost of encoding full-resolution history. Pre-encoding resolution allocation lowers this

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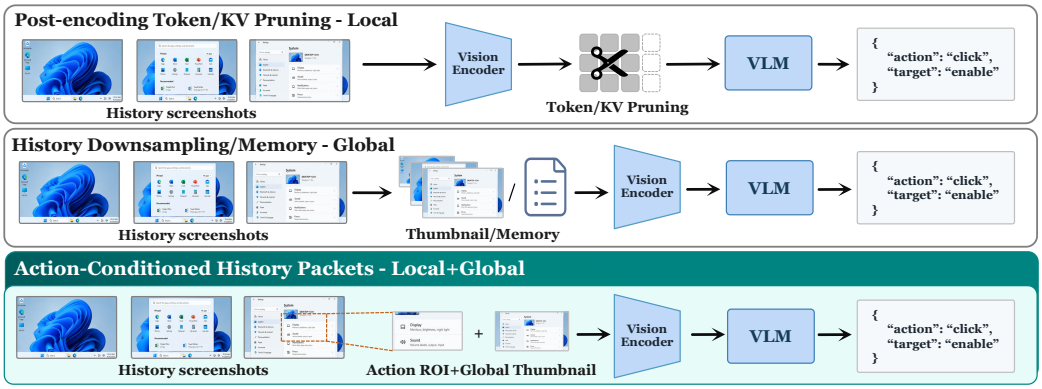


Figure 1: The main idea of Action-Conditioned History Packet. Unlike approaches that optimize a single stage of multi-step inference, our method reduces history redundancy before visual encoding. It represents each historical screen as a History Packet with a global thumbnail and an action-conditioned local crop.

cost by reducing temporally distant frames under a global token budget (Xu et al., 2026). Separately, template-based structured decoding avoids unnecessary autoregressive decisions for fixed action syntax (Jivani et al., 2025; Guidance-AI, 2023; Beurer-Kellner et al., 2024), but not historical visual cost. The open question is whether action-conditioned global–local allocation can improve on temporal-resolution schedules and complement structured output decoding.

Our analysis reveals globally redundant but locally heterogeneous GUI histories: adjacent screens are feature-redundant, whereas their corresponding local regions are not (Figure 2). Teacher-forced masking further shows that similarity alone is insufficient for deletion, as temporally similar history can still alter gold-action likelihood. Qualitative attention maps further show that concentrated history attention can coincide with regions near executed actions (Appendix A). We therefore use past actions as spatial anchors, retaining compact global state and localized interaction-relevant detail.

These observations motivate PACT-GUI, a training-free framework that compresses history before visual encoding using *P*acketized *A*ction-*C*onditioned His*T*ory. Each History Packet combines a low-resolution global thumbnail with a high-resolution crop around an executed action, while retaining the current screen at full resolution, as illustrated in Figure 1. This representation reduces historical pixel processing, visual tokens, and multimodal prefill cost. For end-to-end deployment, it is paired with template-based structured decoding, which generates only task-dependent action fields autoregressively.

To evaluate the effectiveness of our method, we conduct comprehensive experiments across four GUI benchmarks, AndroidControl-Curated, AITW, Mind2Web and GUIOdyssey. PACT-GUI achieves a $6.62\times$ reduction in mean end-to-end computation. In particular, on AndroidControl-Curated, the full PACT-GUI system reduces end-to-end FLOPs by $5.18\times$ and latency by $1.79\times$. Our experiments further separate the effects of the history representation and output protocol, and characterize their quality–efficiency trade-offs.

Our contributions are fourfold:

- We empirically show that multi-step GUI histories are globally redundant but locally heterogeneous, and that visual similarity alone is insufficient for selecting historical content to delete.
- We propose Action-Conditioned History Packets, a training-free, pre-encoding history representation that combines a compact global view with a high-resolution local view anchored by an executed action.
- We evaluate an end-to-end GUI agent acceleration framework that pairs History Packets with a template-based structured decoding protocol and measures their individual and combined quality–efficiency trade-offs.

- We conduct comprehensive experiments across four multi-step GUI tasks to demonstrate a favorable task-quality–efficiency tradeoff: PACT-GUI achieves a $6.62\times$ reduction in mean end-to-end computation.

2 Related Work

Efficient GUI agents for multi-step tasks. Pure-vision GUI agents must process high-resolution observations over long interaction traces. Existing methods fall into three groups. *Learned history representations* encode the original screen stream with a history-aware encoder or resampler (Zhang et al., 2025a; Lu et al., 2025), or compress it through consistency-guided, textual, or learned branches (Chen et al., 2025; Liu et al., 2025; Zhou et al., 2026b). *Training-free acceleration* instead prunes visual tokens after representation construction, compresses KV states, or allocates resolution by recency and structural saliency (Li et al., 2026; Huang et al., 2025; Zhou et al., 2026a; Xu et al., 2026). More generally, long-horizon agents summarize, reflect on, or externalize trajectory state as textual memory (Yao et al., 2023; Shinn et al., 2023; Packer et al., 2023). These approaches either retain the cost of encoding full-resolution history or may discard fine-grained action-local context. In contrast, PACT-GUIconstructs a global thumbnail and an action-conditioned local ROI before visual encoding, retaining page-level state and action-local detail with less visual computation.

Generic visual token pruning for VLMs. Generic VLM pruning usually ranks a flattened visual-token sequence without modeling the temporal role of individual screens. *Post-LLM selection* uses layer-wise or text-relevant attention to rank encoded tokens (Chen et al., 2024; Zhang et al., 2025b), whereas *pre-LLM transformation* selects, merges, or learns compact spatial token sets (Shang et al., 2025; Alvar et al., 2025; Ryoo et al., 2021; Rao et al., 2021; Bolya et al., 2023). Because both operate on a shared encoded token pool, they retain the cost of encoding full-resolution history and do not distinguish page-level state from the local outcome of an earlier interaction. In contrast, PACT-GUIuses the executed action to construct a separate local view and a global thumbnail before visual encoding, while retaining the current screen at full resolution for grounding.

Efficient structured decoding. Structured-generation systems reduce autoregressive work by separating fixed syntax from dynamic slots or enforcing output programs. *Template-based decoding* forwards known spans and predicts only variable slots (Jivani et al., 2025); *program- and grammar-constrained generation* expresses, compiles, or executes output constraints (Guidance-AI, 2023; Beurer-Kellner et al., 2024; Willard & Louf, 2023; Dong et al., 2025; Zheng et al., 2024b). Rather than introducing a new structured-decoding algorithm, PACT-GUIcomplements these approaches by compressing history before visual encoding, thereby accelerating visual encoding, prefilling, and decoding end to end.

3 Preliminary

In this section, we examine how redundancy in visual histories and action outputs contributes to GUI inference cost. Our analysis shows that visually similar screenshots can still affect action prediction and explores how past actions can guide history compression. We also quantify the decoding spent on fixed syntax, linking these observations to the history representation and action-generation strategy introduced in Section 4.

3.1 Problem Formulation

We consider a GUI agent that executes an instruction q sequentially. At step t, it observes the current screenshot X_t and a bounded executed history

$$H_t = \{(X_i, a_i)\}_{i=\max(1,t-K)}^{t-1}, \tag{1}$$

where K is the history-window size and a_i follows X_i. It generates a structured string y_t, parsed as $a_t = \mathrm{Parse}(y_t)$. Our analyses use AndroidControl-Curated (Leung et al., 2025), whose trajectories support real-world mobile UI control (Li et al., 2024a).

At inference, a conventional vision-language agent serializes q, the full-resolution current screen, and every retained screen–action pair into the multimodal prompt

$$P_t = \mathrm{Prompt}\Big(q, X_t, \{(X_i, a_i)\}_{i=\max(1,t-K)}^{t-1}\Big). \tag{2}$$

Let $N(X)$ be the visual-token count of screenshot X. The history contributes

$$N_{\mathrm{hist}}(t) = \sum_{i=\max(1,t-K)}^{t-1} N(X_i) \tag{3}$$

visual tokens at each step, reprocessing historical pixels even when adjacent screens change little. This cost is acute for long-horizon agents, motivating learned screen-stream encoders and history simplification (Zhang et al., 2025a; Chen et al., 2025).

Before generation, the model performs multimodal *prefill* over P_t,

$$(\mathbf{h}_t, \mathcal{K}_t) = \mathrm{Prefill}_\theta(P_t), \tag{4}$$

where $\mathcal{K}_t$ is the generation KV cache. Historical redundancy thus inflates image processing, visual encoding, prefill, and the cache state before decoding—unlike post-encoding token or cache compression, which begins after visual representations are formed (Li et al., 2026; Huang et al., 2025; Zhou et al., 2026a; Zhang et al., 2023; Xiao et al., 2023; Li et al., 2024b; Cai et al., 2024; Liu et al., 2023d; Jiang et al., 2023).

The decoder then generates the complete action string autoregressively,

$$p_\theta(y_t \mid P_t) = \prod_{j=1}^{M_t} p_\theta(y_{t,j} \mid P_t, y_{t,<j}), \tag{5}$$

where M_t is the output length. Each token requires a transformer forward pass that reads and extends $\mathcal{K}_t$, so per-step cost decomposes as

$$C_t = C_{\mathrm{vision}}(X_t, H_t) + C_{\mathrm{prefill}}(P_t) + C_{\mathrm{decode}}(M_t, \mathcal{K}_t). \tag{6}$$

The following analyses motivate compact visual-history packets and contextualize the structured output protocol used in our end-to-end evaluation.

3.2 Historical Redundancy Analysis

Unless stated otherwise, we analyze uncompressed four-frame histories from AndroidControl-Curated using post-merge visual features from Qwen3-VL-4B- Instruct. We sample 874 task-balanced decision points from 3,012 eligible points (seed 42). This intentionally analyzes the original full-history input rather than the proposed packet representation.

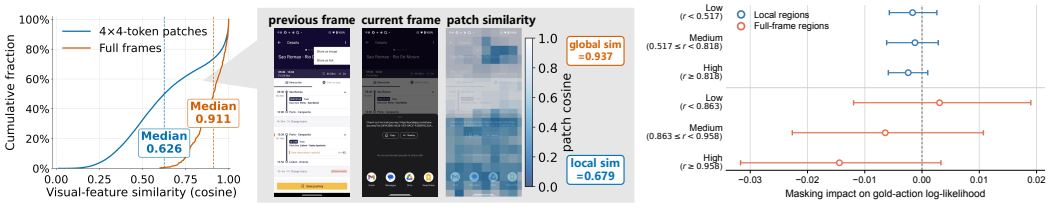
3.2.1 Global Redundancy and Local Heterogeneity

For adjacent historical screens X_{i-1} and X_i, we measure global redundancy using the cosine similarity between their spatially averaged visual features, and local redundancy using corresponding normalized regions r:

$$\begin{aligned} R_{\mathrm{global}}(X_i, X_{i-1}) &= \cos\big(\bar{f}(X_i), \bar{f}(X_{i-1})\big), \\ R_{\mathrm{local}}(r_i, r_{i-1}) &= \cos\big(f(r_i), f(r_{i-1})\big). \end{aligned} \tag{7}$$

Here, $f(\cdot)$ is the mean post-merge feature of a region and $\bar{f}$ is the mean feature of a full screen. Local regions are 4×4 tiles on the final visual-token grid, matched to the same normalized spatial extent in the preceding screen.

Figure 2(a) shows a clear scale-dependent pattern. Full screens have median similarity 0.911 ($n = 2{,}622$), whereas local tiles are substantially more heterogeneous, with median similarity 0.626 ($n = 452{,}898$). This scale-dependent pattern suggests that wholesale frame deletion can lose localized changes. It motivates a multiscale representation: a compact global view can retain page-level context, while a separate local view preserves selected fine-grained evidence.



(a) **(b)**

Figure 2: Preliminary observations on multi-step GUI inference. *(a)* Full history screens are visually redundant, whereas local regions are more heterogeneous, motivating complementary global and local views. *(b)* Even highly redundant regions can affect the gold-action likelihood; visual redundancy alone is therefore insufficient for selecting history to discard.

3.2.2 Visual Redundancy Does Not Imply Decision Irrelevance

Visual similarity alone does not establish that a region is dispensable. We therefore evaluate the local sensitivity of the gold action likelihood. For a historical region r, let $H_t \setminus r$ denote the same prompt in which only r is replaced with the image-wise mean RGB value; the current screen, other history screens, and prompt length remain unchanged. We define

$$\mathcal{L}(H_t) = \sum_j \log p_\theta(y_{t,j} \mid q, X_t, H_t, y_{t,<j}),$$
$$U(r) = \mathcal{L}(H_t) - \mathcal{L}(H_t \setminus r). \tag{8}$$

Positive $U(r)$ indicates that masking r lowers the teacher-forced likelihood of the gold action. This is a local diagnostic of model sensitivity, not an intervention on free-running trajectory success.

Figure 2(b) plots this utility against the redundancy in Equation 7 for 1,600 comparable regions from step-level correct decision points. Utility remains dispersed even among high-similarity regions. Feature redundancy is therefore not a sufficient deletion criterion: temporally similar content can still influence the model's confidence in the correct action. This result argues against simply discarding high-redundancy history tokens and motivates retaining a low-cost global thumbnail together with targeted local evidence.

3.3 Structured Action Protocols as an Output-Side Cost

The visual-history observations address the cost incurred before decoding. GUI actions also contain fixed tags, JSON fields, delimiters, and closing symbols, whereas only action semantics and, when needed, coordinates remain unknown. In our AndroidControl-Curated audit of the ordinary serialized outputs, an average of 39.33 autoregressive decoding steps is used per action, of which approximately 14.07 (35.8%) emit deterministic template syntax rather than task-dependent content. This substantial fixed-syntax share identifies an output-side source of avoidable autoregressive decisions. It motivates the template-based structured decoding protocol used in our end-to-end system, which forwards known syntax in blocks to construct the KV cache and generates only task-dependent slots (Jivani et al., 2025; Guidance-AI, 2023; Beurer-Kellner et al., 2024). The known syntax is still processed by the transformer; the protocol avoids separate autoregressive decisions and format search for these tokens. This complementary observation motivates measuring visual encoding, prefilling, and decoding together in the end-to-end evaluation.

4 Method

PACT-GUI is an end-to-end deployment framework centered on the *Action-Conditioned History Packet*, a training-free representation for historical GUI screens. The packet is constructed before visual encoding and therefore reduces the visual work associated with retained history. To assess the resulting system under a realistic action protocol, we pair History Packets with a template-based structured decoding component, which is complementary to the History Packet method, allowing us to measure visual encoding, prefilling, and decoding in one inference pipeline without changing the underlying VLM or environment action interface. Figure 3 summarizes this end-to-end pipeline.

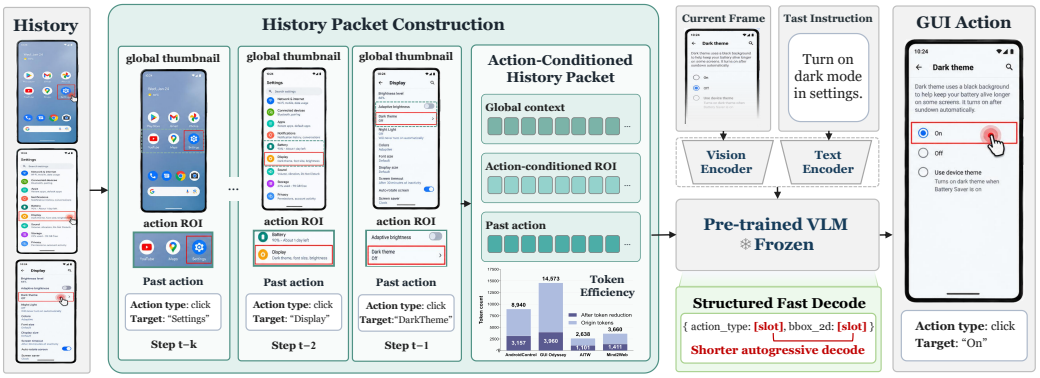


Figure 3: System overview of PACT-GUI. Our main component is the Action-Conditioned History Packet, which rewrites each historical screen before visual encoding as a global–local visual packet. The structured decoding protocol is a component used for end-to-end deployment evaluation.

4.1 System Overview

Given an instruction q, current screen X_t, and an already executed history H_t, PACT-GUI replaces only the historical screenshots with History Packets:

$$\widetilde{H}_t = \left\{(\mathcal{S}_i, a_i)\right\}_{i=\max(1,t-K)}^{t-1}, \qquad \mathcal{S}_i = (G_i, L_i), \tag{9}$$

where G_i and L_i are a global thumbnail and an action-conditioned local view, respectively. The resulting multimodal prompt is

$$\widetilde{P}_t = \mathrm{Prompt}\left(q, X_t, \widetilde{H}_t\right). \tag{10}$$

Crucially, X_t remains at its original input resolution. Thus, the agent can inspect the complete current interface when selecting the next action, while the cost of representing historical screens is reduced before they enter the visual encoder.

After multimodal prefilling of $\widetilde{P}_t$, the system uses the structured output protocol described in §4.3 to serialize its action. This protocol is included to evaluate the complete quality–efficiency trade-off; the following subsection describes the proposed History Packet representation.

4.2 Action-Conditioned History Packet

The observations in §3.2 motivate preserving two complementary forms of historical evidence. Although adjacent GUI screens are often redundant at the page level, localized changes can remain informative. We therefore retain a compact global view of each historical screen and an action-anchored high-resolution view centered on the target of the corresponding action. For each historical screen–action pair (X_i, a_i), we construct the resulting two-scale History Packet:

$$G_i = \mathrm{Resize}(X_i; s_g), \qquad L_i = \mathrm{Resize}(\mathrm{Crop}(X_i, B(a_i)); s_\ell), \qquad \mathcal{S}_i = (G_i, L_i), \tag{11}$$

where s_g and s_ℓ are the global and local image scales. The low-resolution G_i retains compact evidence about page layout, navigation state, and other interface-level context. The high-resolution L_i retains detail around the action target and its neighboring controls, preserving local interaction context relevant to subsequent decisions. In practice, the resized dimensions are aligned with the visual processor's patch grid.

Action-conditioned local view. Local crops should retain context around a past action rather than an arbitrary region. In a qualitative diagnostic, attention from teacher-forced gold-action tokens to historical visual tokens can concentrate in or near elements manipulated at earlier steps (Appendix A). This observation suggests using the executed action as a spatial prior for high-resolution history allocation. We therefore anchor $B(\cdot)$ to the executed action rather than to a fixed center crop.

At deployment, $B(a_i)$ uses only the execution log: click, tap, and long-press actions use the target box or coordinate; scroll and swipe actions use a direction-specific strip; and other actions use a centered fallback. Crops expand to retain local context. Thus, all $B(a_i)$ for $i < t$ are available at step t without accessing the unknown future action a_t.

Let $N(\cdot)$ be the visual-token count defined in §3.1. History Packet construction changes the history budget from

$$\sum_{i=\max(1,t-K)}^{t-1} N(X_i) \quad \longrightarrow \quad \sum_{i=\max(1,t-K)}^{t-1} [N(G_i) + N(L_i)]\,. \tag{12}$$

Because this replacement occurs before visual encoding, it reduces both the pixels processed by the vision module and the visual-token sequence consumed by multimodal prefill. History Packets therefore differ from token or KV-cache pruning methods that select from a representation only after visual content has already been processed.

4.3 Template-Based Structured Output Protocol

GUI actions contain fixed tags, JSON fields, delimiters, and closing symbols, while the action semantics and, when required, bounding-box coordinates remain task dependent. For end-to-end evaluation, we adopt a template-based structured decoding protocol that handles the former as known token blocks and generates only the latter autoregressively. This template–slot execution pattern is closely related to TeCoD (Jivani et al., 2025), which applies partitioned constrained decoding to retrieved SQL templates, and to general structured generation systems (Guidance-AI, 2023; Beurer-Kellner et al., 2024). Here, we use the fixed GUI action schema as an adopted complementary component rather than as a new decoding algorithm. Writing a serialized action as

$$y_t = (d_0, z_1, d_1, \ldots, z_m, d_m), \tag{13}$$

the deterministic spans d_j are forwarded to update the KV cache, whereas the task-dependent slots z_j are generated token by token. Known syntax is thus still processed by the transformer; the protocol avoids separate autoregressive decisions and format search for those tokens.

5 Experiments

5.1 Experimental Setup

Benchmarks and metrics. We evaluate PACT-GUI on the complete evaluation splits of four multi-step GUI benchmarks: AndroidControl-Curated(high split) (Leung et al., 2025), GUIOdyssey (Lu et al., 2025), Android in the Wild (AITW) (Rawles et al., 2023), and Mind2Web (Deng et al., 2023). We follow each benchmark's official step success rate (SR) for AndroidControl-Curated; action match score (AMS) for GUIOdyssey; average action accuracy for AITW; and step success rate for Mind2Web. We report end-to-end FLOPs as a hardware-agnostic efficiency metric; ablations report per-instance end-to-end, prefill, and decode latency.

Model and implementation. All methods use the same Qwen3-VL-4B-Instruct checkpoint (Bai et al., 2025) with deterministic decoding. We retain the current screen uncompressed and provide at most four preceding screens as history. Experiments run as distributed evaluation on seven NVIDIA L20 GPUs. CUDA synchronization is enabled around total inference timing, and all reported results are aggregated over the complete evaluation split of the corresponding benchmark.

Baselines. We compare against the uncompressed Qwen3-VL backbone and eight training-free baselines. GUI-KV (Huang et al., 2025) and ST-Lite (Zhou et al., 2026a) compress the KV cache after visual content has entered the model. HistPrune (Li et al., 2026) and GUIPruner (Xu et al., 2026) are GUI-oriented visual-history or token-pruning baselines. FastV (Chen et al., 2024), SparseVLM (Zhang et al., 2025b), LLaVA-PruMerge (Shang et al., 2025), and DivPrune (Alvar et al., 2025) are generic visual token-pruning methods. For each benchmark, pruning baselines use approximately PACT-GUI-matched overall prompt-token budgets. We therefore compare quality–FLOPs trade-offs attributable to compression timing and location.

Table 1: Quality–efficiency trade-offs on complete benchmark evaluation splits. Each pair reports task quality (↑) and end-to-end FLOPs; the final pair is the unweighted mean. Bold and underlining denote the best and second-best values, respectively. All methods use QWEN3-VL-4B-INSTRUCT; pruning baselines use approximately matched overall prompt-token budgets (details in §5.2).

	AndroidControl-Curated High		GUIOdyssey		AITW		Mind2Web		Average	
Method	SR (↑)	TTFLOPs (↓)	AMS (↑)	TTFLOPs (↓)	Avg. Acc. (↑)	TTFLOPs (↓)	SR (↑)	TTFLOPs (↓)	Acc. (↑)	TTFLOPs (↓)
Qwen3-VL	55.20	288.56	**60.50**	747.76	47.58	48.08	<u>47.09</u>	68.01	<u>52.59</u>	288.10
GUI-specific training-free pruning										
GUI-KV	53.80	181.77	46.98	388.13	47.10	36.41	**47.19**	52.72	48.77	164.76
ST-Lite	54.20	181.73	46.07	700.96	46.82	35.73	45.70	51.78	48.20	242.55
HistPrune	<u>57.00</u>	92.97	46.75	689.01	<u>47.81</u>	19.91	41.22	26.56	48.20	207.11
GUIPruner	54.70	**47.45**	45.85	**67.93**	46.38	**12.74**	42.33	**15.21**	47.32	**35.83**
Generic VLM pruning										
FastV	51.90	236.71	47.70	269.70	46.39	37.04	44.68	35.27	47.67	144.68
SparseVLM	52.00	104.18	45.93	201.73	47.04	24.60	43.19	32.91	47.04	90.86
LLaVA-PruMerge	51.80	175.08	48.39	370.16	47.39	35.80	15.30	27.11	40.72	152.04
DivPrune	43.50	86.83	38.39	163.06	43.70	21.92	16.55	27.32	35.54	74.78
PACT-GUI	**63.90**	<u>55.72</u>	<u>59.36</u>	<u>86.22</u>	**50.95**	<u>14.94</u>	44.44	<u>17.23</u>	**54.66**	<u>43.53</u>

5.2 MAIN COMPARATIVE RESULTS

Table 1 compares quality and end-to-end FLOPs across 4 benchmarks. We draw three observations.

PACT-GUI achieves the strongest average quality among the compared methods. Its unweighted average score is 54.66%, exceeding the uncompressed backbone (52.59%) and every pruning baseline. The gain is driven by two clear wins: on AndroidControl-Curated, PACT-GUI obtains the best SR (63.90%), improving over the uncompressed model by 8.70 points and over the strongest pruning baseline by 6.90 points; on AITW, it reaches the best average action accuracy (50.95%), a 3.14-point improvement over the next-best pruning result (47.81%). These results show that compressing history before visual encoding need not sacrifice, and can improve, next-action prediction.

The History Packet delivers a substantially better accuracy–cost trade-off than full visual history. Relative to the uncompressed backbone, PACT-GUI lowers end-to-end FLOPs from 288.56 to 55.72 TFLOPs on AndroidControl-Curated ($5.18\times$), from 747.76 to 86.22 TFLOPs on GUIOdyssey ($8.67\times$), and from 48.08 to 14.94 TFLOPs on AITW ($3.22\times$), while improving quality on the first and third benchmarks and remaining close to full-history quality on GUIOdyssey (59.36% versus 60.50% AMS). Averaged across benchmarks, PACT-GUI uses 43.53 TFLOPs, $6.62\times$ less than the backbone (288.10 TFLOPs).

PACT-GUI outperforms the only lower-cost pruning baseline across all benchmarks. GUIPruner is the only method with lower FLOPs, but PACT-GUI improves over it by 9.20 SR points on AndroidControl-Curated, 13.51 AMS points on GUIOdyssey, 4.57 average-action-accuracy points on AITW, and 2.11 step- success-rate points on Mind2Web. Thus, under approximately matched prompt-token budgets, PACT-GUI consistently offers a stronger quality–efficiency trade-off than the most aggressively compressed baseline.

Token-budget alignment. Table 1 uses approximately matched overall prompt-token budgets. PACT-GUI's measured retention rates are 35.00% on AndroidControl-Curated, 26.45% on GUIOdyssey, 44.07% on AITW, and 38.57% on Mind2Web; the corresponding overall retention rates configured for every pruning baseline are 35%, 30%, 45%, and 40%, respectively. The small gaps between configured and measured budgets result from discrete visual resizing and variable token counts across examples.

5.3 ABLATION STUDIES

5.3.1 ABLATING THE HISTORY PACKET

Historical visual information is necessary for reliable action prediction. Removing history lowers SR from 55.2% to 49.2%. Text-only history partially recovers performance (57.0% SR), but remains below visual-history variants, showing that the action record alone does not capture all relevant state. Details are reported in Figure 4.

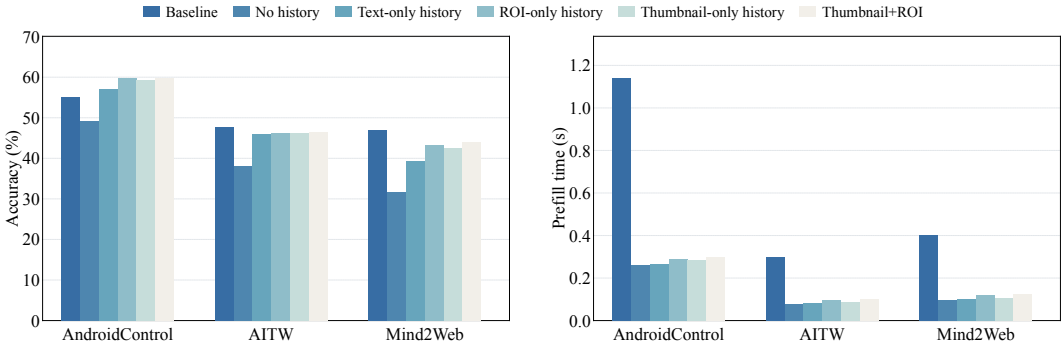


Figure 4: History Packet ablation across AndroidControl, AITW, and Mind2Web. Each dataset group compares full history, history ablations, and the combined thumbnail+ROI representation. The left panel reports task accuracy; the right panel reports per-instance prefill time.

Table 2: Independent and combined effects of History Packets (HP) and structured fast decoding (SFD) on complete evaluation splits. SR denotes step success rate and Avg. Acc. denotes average action accuracy (both ↑). Dashes denote measurements not recorded in the evaluation summary.

	AndroidControl-Curated High					AITW				
Configuration	E2E (s) ↓	Prefill (s) ↓	Decode (s) ↓	Decode steps ↓	SR (%) ↑	E2E (s) ↓	Prefill (s) ↓	Decode (s) ↓	Decode steps ↓	Avg. Acc. (%) ↑
Baseline	5.540	1.141	2.064	39.33	55.20	2.539	0.298	0.977	20.63	47.58
+ SFD	5.348	1.410	1.448	25.26	**64.60**	2.395	0.354	**0.762**	**14.74**	**52.15**
+ HP	3.864	**0.336**	1.947	39.28	59.80	2.17	**0.149**	0.95	20.62	47.08
+ HP + SFD	**3.092**	0.397	**1.299**	**24.67**	63.90	**2.12**	0.182	0.763	15.06	50.95

Global context and action-conditioned local detail are complementary. On Mind2Web, the combined History Packet attains 44.05% accuracy, exceeding ROI-only (43.19%) and thumbnail-only (42.56%) history. On the same benchmark, it reduces prefill time from 0.401 s to 0.126 s ($3.17\times$). Figure 4 shows similarly favorable accuracy–prefill operating points on AndroidControl and AITW,

5.3.2 Disentangling Visual Compression and Structured Decoding

History Packets and structured decoding accelerate different stages. On AndroidControl-Curated, History Packets principally reduce prefill time (1.141 s to 0.336 s) without lalargely rgely reducing autoregressive steps. Structured decoding instead reduces decoding steps (39.33 to 25.26) and decode time (2.064 s to 1.448 s). The same pattern holds on AITW: History Packets lower prefill time, whereas structured decoding lowers decoding time and steps. Details are reported in Table 2.

Two components offer complementary efficiency–quality effects. Their combination attains the lowest AndroidControl-Curated end-to-end time (3.092 s). HP largely preserves task quality, raising AndroidControl-Curated SR from 55.20% to 59.80% and changing AITW average accuracy by only -0.50 points. In contrast, SFD's stronger template constraints improve accuracy on both benchmarks (to 64.60% SR and 52.15% average accuracy, respectively).

6 Conclusion

PACT-GUI reduces multi-step GUI inference cost by replacing each historical screen, before visual encoding, with an Action-Conditioned History Packet: a global thumbnail and an action-anchored high-resolution ROI. This global–local representation preserves compact interface context and interaction-local evidence; paired with a structured decoding protocol, it reduces visual encoding, prefilling, and decoding cost without training. Across four complete evaluation splits, PACT-GUI achieves the best average quality among compared methods and a $6.62\times$ reduction in mean end-to-end FLOPs relative to full-history Qwen3-VL. These results support action-anchored, pre-encoding history allocation for efficient GUI agents.

AI USE STATEMENT

Generative AI tools were used as editorial aids during manuscript preparation, including language polishing, presentation refinement, and LaTeX typesetting. The authors reviewed and revised all AI-assisted material, verifying that the manuscript faithfully reports the proposed method and experimental findings. The authors retain sole responsibility for the scientific content, accuracy, and conclusions of this work.

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A Qualitative History-Attention Diagnostic

This diagnostic examines whether the model's read path visually aligns with historical interaction locations. We use the diagnostic only to motivate the action-conditioned allocation in §4.2; the masking analysis in §3.2.2 and the History Packet ablations provide the corresponding utility and end-task evidence.

Experimental design. For a full-resolution history prompt, we append the ground-truth serialized action and use teacher forcing. Let $A_{l,h,q,k}$ denote the attention from a gold-action query token q at decoder layer l and head h to historical visual token k. To visualize spatial allocation within history, we normalize attention over all historical visual tokens and average the resulting maps over the selected layers, heads, and gold-action query tokens. This produces one heatmap per historical screen. The diagnostic recomputes only these gold-query-to-history attention rows while leaving the model's standard SDPA forward pass intact; it neither crops nor prunes tokens, and does not alter the benchmark inference path.

Setting. We run the diagnostic on randomly selected task-balanced decision points from AndroidControl-Curated High using Qwen3-VL-4B-Instruct, seed 42, and four full-resolution historical screens per prompt. We use the final third of decoder layers and all tokens in the gold action as queries. Figure 5 shows four representative cases and, for compactness, their three most recent historical screens ($t-3$ to $t-1$). Green boxes mark the tight ground-truth target box (`gt_min_bbox`) of the action executed at each historical step; they are visualization annotations rather than the packet crop boxes used at deployment.

Qualitative observations. Across the displayed examples, high-attention patches frequently coincide with or lie close to the annotated control that was manipulated in the corresponding historical state—for example, interface controls in cases (a)–(c) and the interaction area in case (d). Attention is not confined to the target box: nearby controls, text, and layout context can also receive mass. This pattern supports preserving a context-expanded local view rather than a point-sized crop. It also motivates pairing the local view with a global thumbnail, since the maps remain spatially distributed in several screens. These maps provide qualitative evidence only.

B Benchmark Protocols and Evaluation Settings

Overview. We evaluate on four multi-step GUI benchmarks spanning mobile and web interaction: AndroidControl-Curated (Leung et al., 2025), Android in the Wild (AITW) (Rawles et al., 2023), Mind2Web (Deng et al., 2023), and GUIOdyssey (Lu et al., 2025). At each decision point, the current screen is retained at its original resolution, and the model receives at most four preceding screens. Unless stated otherwise, historical actions are ground-truth actions under teacher forcing rather than the model's previous predictions.

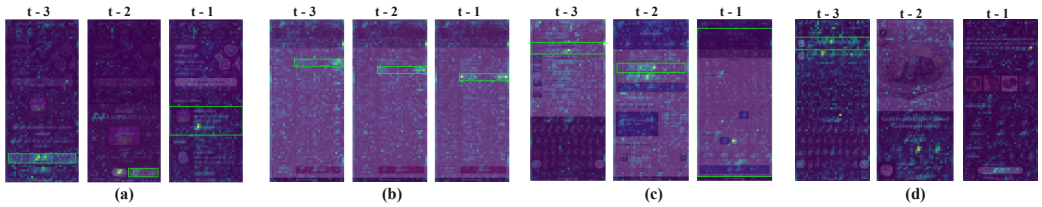


Figure 5: Qualitative history-attention diagnostic on AndroidControl-Curated High. Each group is one decision point; columns show its three most recent full-resolution history screens. Heatmaps visualize attention from teacher-forced gold-action tokens to historical visual tokens, normalized within the retained history and averaged across the final third of decoder layers and all heads. Green boxes denote tight historical action-target annotations (`gt_min_bbox`). Attention can concentrate at or near prior interaction locations, motivating action-conditioned local preservation; it is not a causal importance measure.

Table 3: Overall prompt-token retention in budget-matched comparisons.

Benchmark	Pruning baselines	PACT-GUI
AndroidControl-Curated	35%	35.00%
GUIOdyssey	30%	26.45%
AITW	45%	44.07%
Mind2Web	40%	38.57%

AndroidControl-Curated. We use the `AndroidControl_Curated_High_Task_Improved` split. The model predicts an action type and coordinates; coordinates generated on the 0–1000 scale are mapped back to the current image resolution for evaluation. A prediction is accepted if it matches either the primary ground-truth action or one of the annotated candidate actions. We report step success rate (SR) which requires both the action type and location to be correct.

AITW. We evaluate the `AITW_all` test split, which covers the `general`, `single`, `webshopping`, `install`, and `googleapps` categories and expands each trajectory into action steps. For taps, a prediction is correct when it hits an admissible target box or lies within the normalized distance tolerance of the ground-truth point; text actions require token F1 of at least 0.5, while scroll and drag actions require the correct type and direction. We report average action accuracy for experiments.

Mind2Web. We use `Mind2Web_test_task` and evaluate every action step in each web trajectory. The model outputs a Python dictionary with action coordinates on a 0–1000 normalized scale. An action must have the correct operation type; element correctness requires the predicted point to fall within the annotated bounding box, and TYPE and SELECT actions additionally evaluate the action value through operation F1. We report step success rate for experiments.

GUIOdyssey. We use the `GUIOdyssey_high_task_split`. The model produces a single serialized action command. CLICK and LONG_PRESS actions must have the correct type and either hit the SAM2 target box or satisfy the normalized coordinate-distance tolerance; TYPE actions require Levenshtein similarity of at least 0.5, and SCROLL actions require the correct direction. This split uses macro evaluation: action match score (AMS), which measures step-level action matching.

C Packet Construction and Budget Matching

History Packet construction. We retain at most four prior screens, representing each with one global thumbnail and one action-conditioned local ROI. The thumbnail has a maximum side length of 192 pixels. The ROI has a maximum side length of 224 pixels, a minimum side length of 160 pixels, and a short-side ratio of 0.22. Its anchor is derived only from the already executed historical action: we use the target box or coordinate for click, tap, and long-press actions; a direction-specific strip for scroll and swipe actions; and a centered crop as a fallback. Resize dimensions are aligned with patch size 16 and merge size 2, so the realized visual-token count can differ slightly from that implied by the nominal pixel dimensions.

Token-budget-matched comparisons. We match methods by configured overall token retention, and follow their native policies which may compress the current screen, history, or both.

Table 3 compares baseline targets with PACT-GUI's measured retention; minor gaps reflect image-dependent tokenization and grid-aligned resizing. Because PACT-GUI never retains more tokens, any quality gain is achieved under an equal or stricter budget.